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Probabilistic Combinatorics SS24

Sheet 3

Due: 6. May at 19:00

(1) There are 799 teams in a tournament. Here every team plays against every other team exactly once and every match is decisive, i.e. there are no ties. Show that there are 2 distinct sets A and B of 7 teams each, such that every team of A has won against every team of B.

(2) A proper vertex coloring of an undirected graph $G = (V, E)$ with a set of colors F is a map $f : V \longrightarrow F$, such that for every edge $e \in E$ we have $\#f(e) = 1$, i.e. no adjacent vertices are colored the same way.
Let $G = ([n], E)$ be a bipartite graph and F a finite set of colors. Let $F_1, \dots F_n \subseteq F$, such that $\#F_i \geq \log_2(n)$ for $1 \leq i \leq n$. Show: There is a proper vertex coloring of G such that $f(i) \in F_i$ for every $1 \leq i \leq n$.

(3) There is a chess tournament with 40 players. There are 80 matches played and no two players play against each other more than once. Show: There are at least 5 players, where no two of them play against each other.
Tipp: choose and change.

(4) Let $(a_n)_{n\in\mathbb{N}}$ and $(b_n)_{n\in\mathbb{N}}$ be sequences of real numbers. Decide which of the following statements is always true, which might occur and which can't be true. Give a corresponding proof (for "is always true" or "can't be true") or examples (one example for "holds" and for "doesnt hold" for each "might occur").

(i) $a_n = O(a_n^2)$.

(ii) $a_n \neq O(b_n)$ and $b_n \neq O(a_n)$.

(iii) $a_n = O(b_n)$ and $b_n = o(a_n)$.